HUANCONG GUAN

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RESEARCH INTERESTS

3D Computer Vision, 3D Gaussian Splatting, 3D Scene Understanding

EDUCATION

Shenzhen University (SZU), Shenzhen, China — Sep 2024 – Present
M.Sc. in Computer Science and Technology, College of Artificial Intelligence — GPA: 3.42/4.00

Guangdong University of Technology (GDUT), Guangzhou, China — Sep 2020 – Jun 2024
B.Eng. in Internet of Things Engineering, School of Automation — GPA: 3.75/5.00

PUBLICATION

Robust Instance Segmentation via 3D Gaussian Denoising Constraints
Huancong Guan, Zehai Niu, Jiayi Lyu, Teng Wang, Bin Huang, Linlin Shen, Jinbao Wang
IEEE Transactions on Image Processing (TIP), Accepted

RESEARCH EXPERIENCE

Robust Instance Segmentation via 3D Gaussian Denoising Constraints — Sep 2025 – Aug 2026
Shenzhen University | Advisor: Dr. Jinbao Wang

- Proposed DenoGS, a global-local denoising framework for robust 3D Gaussian Splatting instance segmentation under noisy and cross-view inconsistent 2D supervision.
- Designed prototype-based confidence estimation and spatial filtering to identify reliable supervision, and extended the framework to language-guided 3D instance retrieval.
- Achieved 0.82 mIoU and 0.89 F1 across 41 challenging scenes, with consistent improvements over strong baselines under sparse-view settings.

Semantic-Guided Object Editing in 3D Gaussian Splatting — May 2026 – Present
Shenzhen University | Advisor: Dr. Jinbao Wang

- Designed a remove-repair-restore pipeline with Qwen-based inpainting and multi-view reconstruction to achieve consistent 3D scene editing with reduced artifacts.
- Integrated semantic features with local geometric connectivity, color consistency, and multi-view visibility constraints to achieve accurate Gaussian selection while suppressing background and surface contamination.

Multimodal Classification of Human Oocyte Development — Dec 2025 – Present
Shenzhen University & Shenzhen Maternity and Child Healthcare Hospital

- Proposed the first multimodal framework for predicting embryo euploidy versus aneuploidy, using a GRU-based video encoder for temporal representation learning and an MLP-based encoder for integrating tabular features.
- Achieved 77% test accuracy on a dataset of approximately 577 samples.

AWARDS AND HONORS

National Scholarship for Master's Students, Shenzhen University — 2026
Special Graduate Scholarship, Shenzhen University — 2024
Second-Class University Scholarship, Guangdong University of Technology — 2021, 2023
Second Prize, China Undergraduate Mathematical Contest in Modeling, Guangdong Province — 2022
Third Prize, China Electrical Engineering Mathematical Modeling Competition — 2022

SKILLS

AI Systems: Large Language Models (LLMs), Agent Frameworks, Retrieval-Augmented Generation (RAG)
Programming Languages: Python, C++
Deep Learning: PyTorch, CNN, Transformer, GRU
Languages: Chinese (Native), English (CET-4, CET-6)